
A Stochastic Dynamical Theory of LLM Self-Adversariality: Modeling Severity Drift as a Critical Process

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Abstract

This paper introduces a continuous-time *stochastic dynamical* framework for understanding how large language models (LLMs) may *self-amplify* latent biases or toxicity through their own chain-of-thought reasoning. The model posits an instantaneous “severity” variable $x(t) \in [0, 1]$ evolving under a stochastic differential equation (SDE) with a drift term $\mu(x)$ and diffusion $\sigma(x)$. Crucially, such a process can be consistently analyzed via the Fokker–Planck approach if each incremental step behaves nearly Markovian in severity space. The analysis investigates *critical phenomena*, showing that certain parameter regimes create phase transitions from subcritical (self-correcting) to supercritical (runaway severity). The paper derives stationary distributions, first-passage times to harmful thresholds, and scaling laws near critical points. Finally, it highlights implications for *agents* and extended LLM reasoning models: in principle, these equations might serve as a basis for *formal verification* of whether a model remains stable or propagates bias over repeated inferences.

1 Introduction

When a large language model (LLM) produces text, it conditions each new token on prior tokens, effectively referencing its own chain-of-thought (CoT). Such an iterative, self-referential mechanism can be beneficial—improving reasoning [1]—but it may also *amplify* latent misalignment. Even in the absence of explicit adversarial prompts, once a partial bias or toxic statement appears, subsequent reasoning steps can elaborate and intensify that negativity. This phenomenon can be called *self-adversarial* escalation. Recent empirical work by Shaikh et Al. [5] demonstrates this risk concretely: they found that zero-shot CoT reasoning significantly increases the likelihood of harmful or biased outputs across multiple sensitive domains, with the effect becoming more pronounced in larger models.

While stepwise or discrete Markovian toy models have offered an initial conceptual lens, this paper proposes a **continuous-time stochastic differential equation (SDE)** approach that captures:

- A *drift term* $\mu(x)$ that encodes deterministic escalation or correction of severity.
- A *diffusion term* $\sigma(x)$ capturing the inherent randomness in LLM sampling.
- **Phase transitions** wherein a small parameter change can push the system from subcritical (stable near $x = 0$) to supercritical (runaway severity near $x = 1$).

Why Fokker–Planck? In practice, each short time increment Δt might correspond to generating a small batch of tokens in the chain-of-thought. If the severity $x(t + \Delta t)$ depends only on (1) the current severity $x(t)$ and (2) a well-defined noise process (stemming from the LLM’s sampling randomness), then the *one-step* transition is approximately Markov in x . Such memoryless behavior

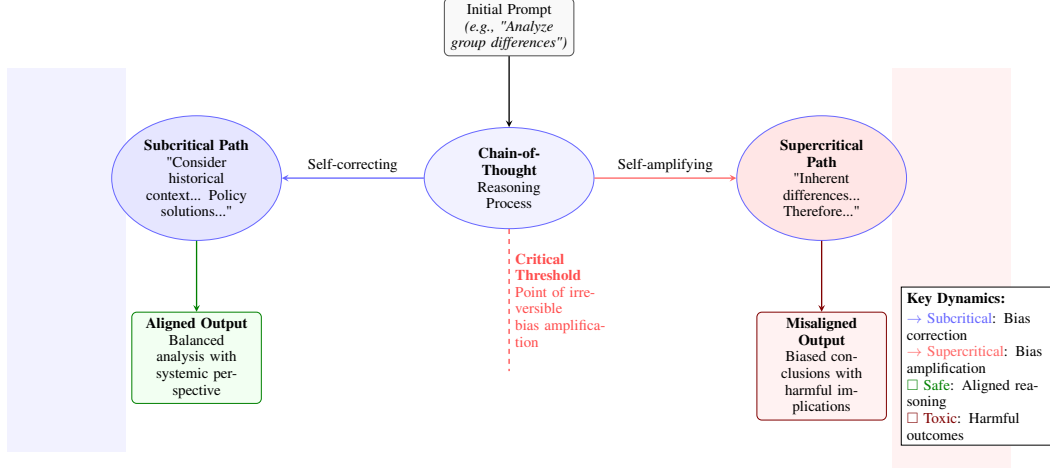


Figure 1: Conceptual diagram of self-amplifying bias in LLM chain-of-thought reasoning. Starting from a neutral prompt, the reasoning process can follow either a subcritical path (where biases are corrected) or a supercritical path (where biases amplify). The critical threshold marks where bias amplification becomes irreversible, leading to divergent outcomes in terms of alignment.

is imperfect but can hold if the relevant context about bias or negativity can be compressed into the scalar severity variable. When the time- (or step-) spacing Δt is small, the transitions can be recast into an SDE limit, making the *Fokker-Planck equation* an apt tool for analyzing the probability flow in severity space.

2 Continuous-Time Severity Model

2.1 State Variable and SDE

Let $x(t) \in [0, 1]$ represent the *instantaneous severity* (e.g., toxicity, bias level) of the LLM’s chain-of-thought at continuous time t . The simplest assumption is

$$dx(t) = \mu(x(t)) dt + \sigma(x(t)) dW(t), \quad (1)$$

- $\mu(x) : [0, 1] \rightarrow \mathbb{R}$ is the *drift function*, capturing deterministic tendencies for severity to grow or diminish.
- $\sigma(x) : [0, 1] \rightarrow \mathbb{R}_{\geq 0}$ is the *diffusion function*, capturing noise intensity at severity level x .
- $W(t)$ is a standard Wiener process (Brownian motion).
- We assume boundary conditions that keep $x(t) \in [0, 1]$ (e.g., reflecting boundaries or saturating behavior).

Remark 1 (Interpretation). *In practice, one might discretize at time intervals Δt , obtaining $x_{t+\Delta t} - x_t \approx \mu(x_t)\Delta t + \sigma(x_t)\sqrt{\Delta t}\eta_t$, where $\eta_t \sim \mathcal{N}(0, 1)$. This approximates the simpler Markov chain model but with continuous-state transitions and explicit parametric forms of drift/diffusion.*

2.2 Approximate Markov Assumption

Why is $x(t)$ Markov? If we measure severity at discrete intervals Δt , then $x(t + \Delta t)$ is not strictly memoryless, since the LLM can reference text from multiple earlier steps. However, if (a) severity $x(t)$ effectively summarizes the "bias content" carried over from prior tokens, and (b) the random generation of new tokens is conditionally independent (beyond severity), the system can be *approximately* Markov at the level of $x(t)$.

Inference for Fokker-Planck. In the limit $\Delta t \rightarrow 0$, standard diffusion-limit arguments [2] show that a well-defined drift $\mu(x)$ and diffusion $\sigma^2(x)$ yields a continuous-state Markov process. The corresponding *Fokker-Planck* partial differential equation (PDE) then emerges naturally to describe

the time evolution of the probability density $P(x, t)$ over severity. This, in essence, *justifies* employing the continuum approach: if each short token block updates x in a manner akin to a small Markov jump, we move to SDE as $\Delta t \rightarrow 0$.

3 Drift and Noise Terms

Drift $\mu(x)$. The drift function is a phenomenological model capturing three key dynamics of severity evolution:

$$\mu(x) = \underbrace{\alpha x(1-x)}_{\text{self-reinforcement}} - \underbrace{\beta x^2}_{\text{alignment}} + \underbrace{\gamma}_{\text{baseline}},$$

where:

- (i) The logistic term $\alpha x(1-x)$ models self-reinforcing bias, borrowed from population dynamics, where severity grows but saturates as $x \rightarrow 1$. Parameter α reflects the model’s tendency to elaborate on and amplify existing biases.
- (ii) The quadratic damping term $-\beta x^2$ represents alignment efforts (e.g., RLHF training) that counteract severity more strongly at higher x . Parameter β quantifies the strength of bias suppression.
- (iii) The constant term $\gamma \geq 0$ captures spontaneous bias emergence from pretraining data or architecture, independent of prior reasoning steps.

The equation admits closed-form solutions and exhibits critical phenomena analogous to phase transitions in physics.

Potential and Critical Behavior. The corresponding potential function $V(x)$ is obtained by integrating $-\mu(x)$. When $\alpha > \beta$, the drift remains positive above a critical threshold $x_c = \frac{\alpha-\beta}{\alpha+\beta}$, leading to supercritical (runaway) behavior. Conversely, when $\beta > \alpha$, the system remains subcritical, with severity returning to low values.

Supercritical vs. Subcritical. If α dominates β in some range, the drift remains positive, causing $x(t)$ to *increase* on average. Conversely, large β ensures $x(t)$ gets pulled back to a stable equilibrium near 0. The first scenario is called "supercritical" and the second "subcritical."

Diffusion $\sigma(x)$. We can let $\sigma(x) = \sigma_0 + \sigma_1 x$ (with $\sigma_0, \sigma_1 \geq 0$), so that the process becomes more volatile at higher severity. This choice reflects the intuition that *controversial or negative* lines of reasoning produce more varied or explosive expansions from the LLM.

4 Fokker–Planck Equation and Stationary Behavior

The Fokker–Planck (FP) equation [3] for the probability density $P(x, t)$ of $x(t)$ is:

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x} [\mu(x) P] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2(x) P]. \quad (2)$$

Intuitive meaning:

- The term $-\frac{\partial}{\partial x} [\mu(x) P]$ captures the *deterministic flow* in severity space.
- The second term $\frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma^2(x) P]$ captures *random spreading*.

4.1 When is Fokker–Planck Valid?

Because $x(t)$ is a (*near*) Markov process in continuous state, it satisfies an SDE of the form (1). The associated Kolmogorov forward equation is precisely (2), known as the Fokker–Planck equation in the physics literature [2]. Hence, if we accept the Markov approximation from Section 2.2, the FP approach is the correct PDE for describing $P(x, t)$.